Physical Activity & BMI Percentile

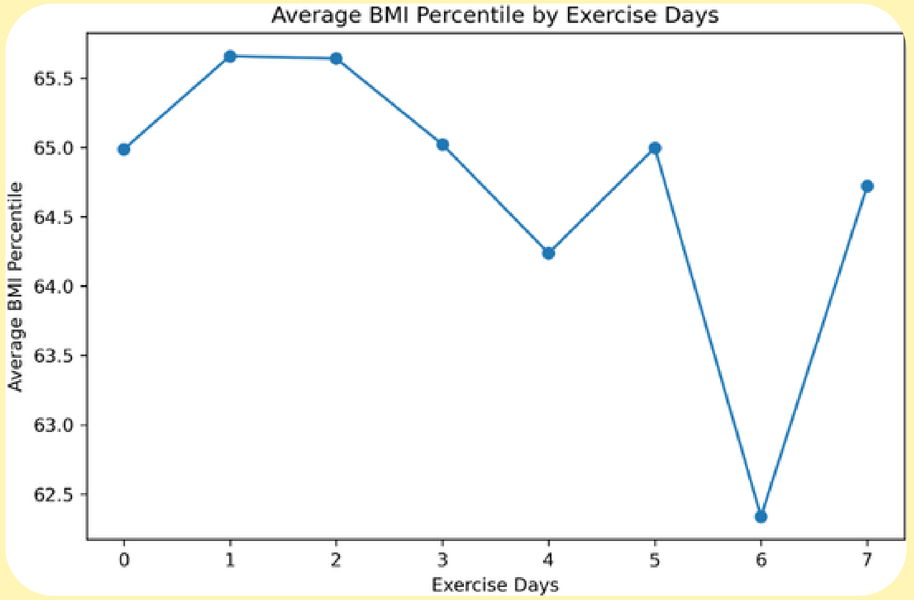
Research Question

Does physical activity affect BMI percentile among high school students?

Method

- Descriptive Satistics
- Correlation Analysis
- Linear Regression
- Gender Comparison Analysis (Extended)

Exercise & BMI



Variable

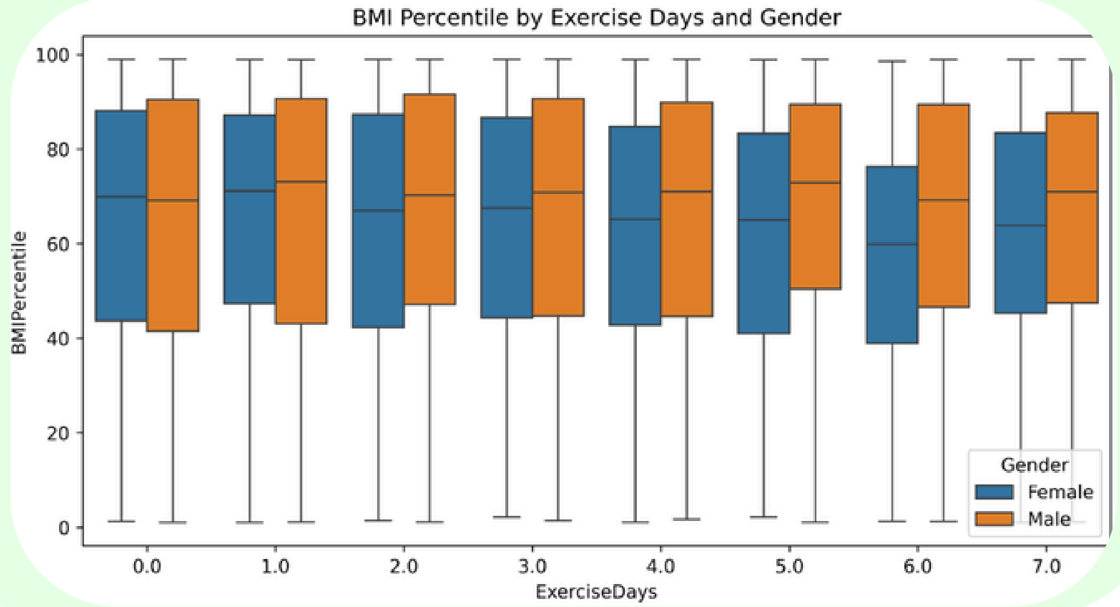
Exercise Days(0–7)	Independent
Gender (Male/Female)	Independent
BMI Percentile(%)	Dependent

Key Result

n=12,527 r=-0.013 Slope=-0.14

R^2=0.0002 P-value=0.145

Regression Equation
BMIPercentile=65.2935-0.1393×ExerciseDays
Exercise frequency explains almost none of the variation in BMI percentile



Interpretation

- Students who exercised more frequently tended to have slightly lower BMI percentile values
- However, the relationship was extremely weak and not statistically significant
- Gender-based analysis showed slight differences in BMI percentile distributions between male and female students
- Gender may influence the relationship between physical activity and BMI percentile

Conclusion

- No significant relationship was found between physical activity and BMI percentile.
- Small gender differences were observed in BMI percentile distributions.
- Gender may influence the relationship between exercise and BMI.